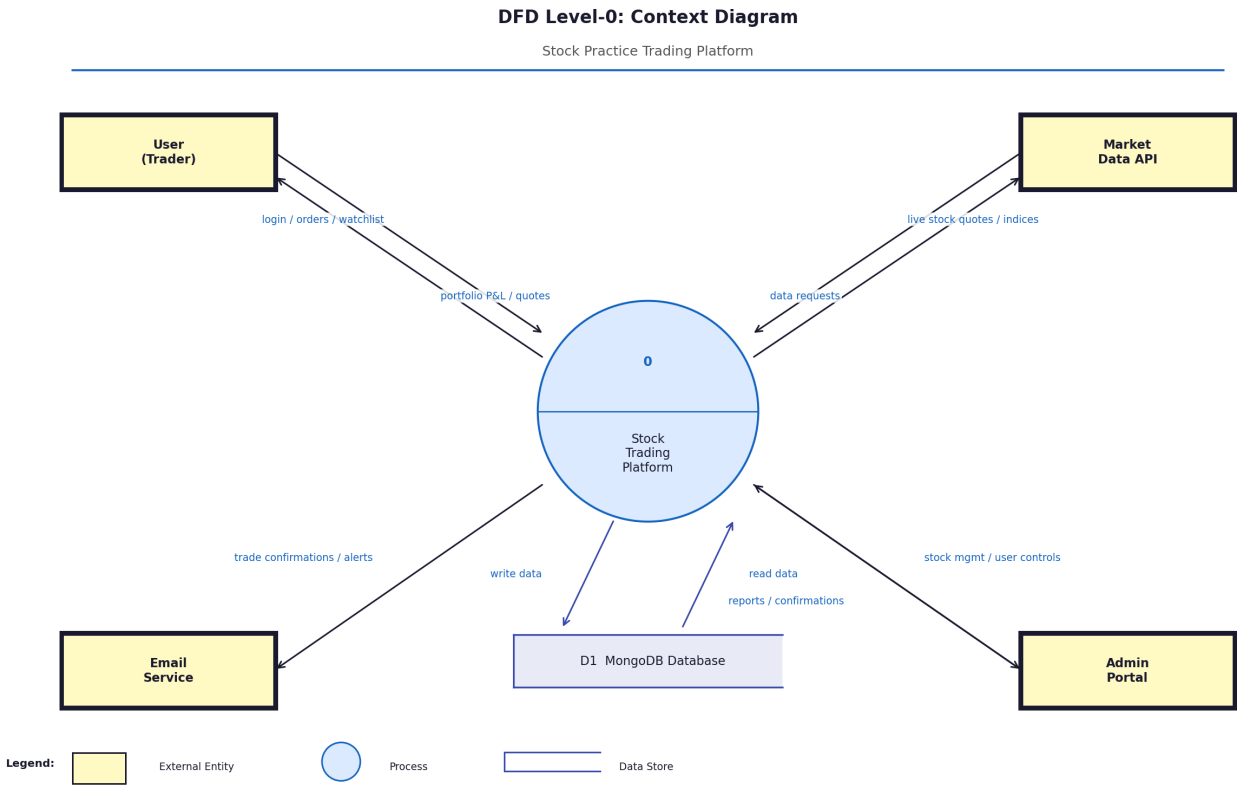
Project Design Phase-II

Data Flow Diagram & User Stories

Date	31 January 2025
Team ID	STK-2025-01
Project Name	Full-Stack Stock Practice Trading Simulation Platform
Maximum Marks	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) visually represents how information flows within the Stock Practice Trading Platform. Level-0 shows the system as a single process with external entities. Level-1 expands the platform into its seven core sub-processes.

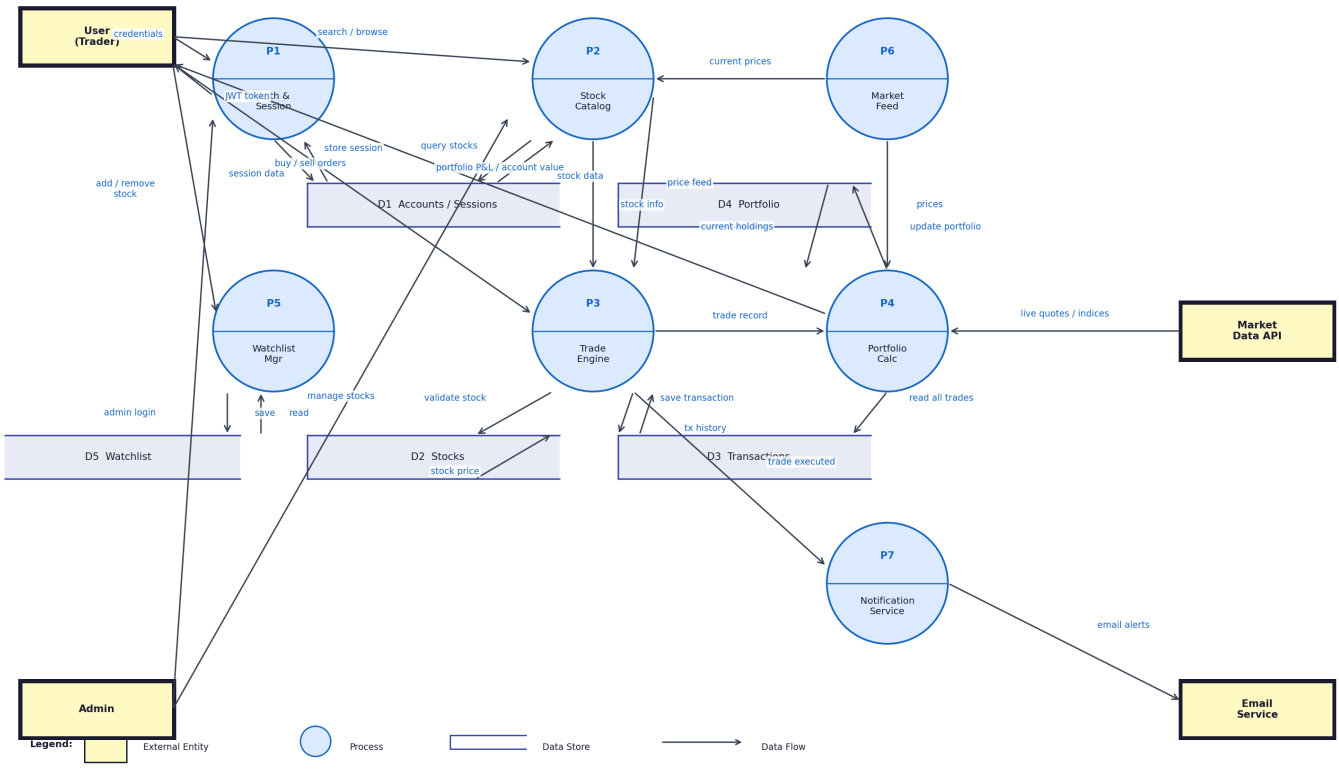


DFD Level-1: Expanded Platform Process View

Level-1 decomposes the central platform process into seven sub-processes: P1 Auth & Session, P2 Stock Catalog, P3 Trade Engine, P4 Portfolio Calculator, P5 Watchlist Manager, P6 Market Feed, and P7 Notification Service — each interacting with the relevant data stores and external entities.

DFD Level-1: Expanded Platform Processes

Stock Practice Trading Platform — Internal Data Flows



User Type	Epic / Functional Area	Story #	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Trader)	Registration	US-01	As a trader, I can register with email and password to access the platform.	Account created with $10,000 virtual balance; redirected to dashboard.	High	Sprint-1
		US-02	As a trader, I receive a verification email after registration to activate my account.	Email arrives within 60 s; clicking the link activates the account.	High	Sprint-1
		US-03	As a trader, I can sign up via Google OAuth for faster registration.	One-click Google signup creates account and opens onboarding wizard.	Medium	Sprint-2
	Login / Session	US-04	As a trader, I can log in with email and password to reach my portfolio.	Valid credentials issue JWT + refresh cookie; invalid credentials show error message.	High	Sprint-1
		US-05	As a trader, I can log out from one or all devices to secure my account.	All session tokens revoked; subsequent API calls return 401 Unauthorized.	High	Sprint-1
	Stock Catalog	US-06	As a trader, I can browse and search available stocks to find ones to trade.	Paginated stock list with live quotes loads in < 500 ms; search filters by name or ticker.	High	Sprint-1
		US-07	As a trader, I can view a stock detail page showing its price chart and OHLCV data.	Detail page renders interactive chart with 30 days of OHLCV history.	High	Sprint-2
	Watchlist	US-08	As a trader, I can add a stock to my watchlist with an optional personal note.	Stock appears in watchlist; note saved; duplicate entries rejected with 409 Conflict.	Medium	Sprint-2
		US-09	As a trader, I can remove a stock from my watchlist at any time.	Stock no longer appears in watchlist after removal; operation is immediate.	Medium	Sprint-2
	Trading	US-10	As a trader, I can place a buy order for a stock if I have sufficient virtual cash.	Transaction created; portfolio cash decreases; holding quantity increases accordingly.	High	Sprint-1
		US-11	As a trader, I can place a sell order for a holding I currently own.	Transaction created; holding reduced; cash increases; realised P&L calculated correctly.	High	Sprint-1
		US-12	As a trader, I see a clear error if I try to buy more than my available cash allows.	Order rejected with 'Insufficient funds' message; portfolio remains unchanged.	High	Sprint-1
		US-13	As a trader, I can edit or cancel a pending transaction before it is executed.	Edited transaction updates portfolio correctly; cancellation removes the record.	Medium	Sprint-2
	Portfolio	US-14	As a trader, I can view my portfolio showing all holdings, unrealised P&L, and total value.	Portfolio recalculates from transaction history after every trade automatically.	High	Sprint-1
		US-15	As a trader, I can view my full transaction history with date and stock filters.	Paginated list shows all trades; date range and stock symbol filters work correctly.	Medium	Sprint-2

User Stories (continued)

User Type	Epic / Functional Area	Story #	User Story / Task	Acceptance Criteria	Priority	Release
	Market Feed	**US-16**	As a trader, I can see near-real-time market index data (S&P 500, Dow, Nasdaq) on my dashboard.	Indices refresh every 60 s; fallback simulated feed activates within 5 s if API fails.	**High**	**Sprint-2**
Admin	**Stock Management**	**US-17**	As an admin, I can create a new stock record in the catalog.	Stock saved with unique symbol; immediately visible to all traders in catalog.	**High**	**Sprint-1**
		US-18	As an admin, I can deactivate a stock so traders cannot place new buy orders for it.	Stock status set to inactive; new buy orders for that stock rejected with 403.	**Medium**	**Sprint-2**
	User Management	**US-19**	As an admin, I can deactivate a trader account to block platform access.	Account inactive; login returns 403 Forbidden; all existing sessions revoked.	**High**	**Sprint-2**
		US-20	As an admin, I cannot deactivate the last active admin account (lockout protection).	System blocks the action with a 409 Conflict error and descriptive message.	**High**	**Sprint-1**
System	**Notifications**	**US-21**	As the system, I send a trade confirmation email when a buy or sell order is executed.	Email with trade summary delivered to user within 2 minutes of execution.	**Medium**	**Sprint-2**
		US-22	As the system, I send a daily portfolio performance summary email to active users.	Email sent at 08:00 AM with P&L delta, top holding, and link to dashboard.	**Low**	**Sprint-3**
	Reliability	**US-23**	As the system, I serve simulated market prices when the external price API is unavailable.	Fallback activates in < 5 s; UI shows 'Simulated Data' badge clearly to the user.	**High**	**Sprint-2**

Priority: ■ High ■ Medium ■ Low